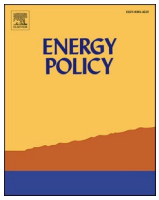
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Navigating trade barriers: How anti-dumping and countervailing policies affect outward foreign direct investment of Chinese solar photovoltaic firms

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ABSTRACT

In the context of rising global trade protectionism, Chinese photovoltaic (PV) firms face increasing anti-dumping and countervailing (ADCV) trade barriers. Outward foreign direct investment (OFDI) has become a crucial strategy to overcome these barriers and expand into overseas markets. Using data from Chinese listed firms between 2007 and 2016, this study empirically examines the impact of ADCV policies on PV firms' OFDI using a multi-period difference-in-differences (DID) approach. The findings show that (1) ADCV measures significantly stimulate OFDI, confirming the "crossing trade barriers effect"; (2) Mechanism analysis identifies three pathways: export barriers push PV firms to use OFDI to bypass trade restrictions; ADCV policies increase market concentration, with stronger firms leveraging OFDI to mitigate negative effects, while weaker firms reduce market share or exit; and PV firms use greenfield investments and cross-border M&A to overcome trade barriers, with M&A being a more efficient strategy; (3) Heterogeneity analysis reveals that firms with higher productivity, those more dependent on external markets, eastern enterprises, and firms located in Belt and Road provinces are more likely to enhance global competitiveness via OFDI. This study offers insights into how Chinese PV firms can navigate trade barriers and provides policy recommendations to optimize internationalization strategies.

1. Introduction

Although the World Trade Organization (WTO) has long advocated for trade liberalization (Howse and Langille, 2023; Baldwin, 2016), recent years have witnessed a resurgence of trade protectionism, especially in strategic industries such as renewable energy. Amid intensifying geopolitical tensions and rising global competition, China's photovoltaic (PV) industry—despite being a global leader—has increasingly become the target of trade barriers. In particular, anti-dumping and countervailing (ADCV) measures have significantly restricted PV exports from China, leading to heightened market uncertainty and escalating external risks (Zhao et al., 2024; Sattich et al., 2021; Zhu et al., 2024). These trade frictions not only obstruct exports at the macro level but also force firms to reconfigure their international strategies, including market selection, supply chain adjustment, and investment decisions (Hajdukiewicz and Pera, 2020; Sun et al., 2014). Therefore, studying the impact of ADCV on the OFDI of Chinese PV firms has become a topic of great practical significance. In the context of escalating trade protectionism, exploring how PV enterprises utilize OFDI to overcome external shocks is essential for ensuring the sustainable development of the industry.

Despite growing attention to trade protectionism, current literature lacks a systematic investigation into how ADCV measures affect the outward investment behavior of Chinese PV firms. Most existing studies focus on the trade-restricting effects of ADCV or examine firm responses at an aggregate level, without isolating the specific reactions of the PV sector—an industry uniquely characterized by high export dependence, technological intensity, and policy sensitivity. Given that the PV industry is central to China's green development agenda and global low-carbon transition, understanding how these firms strategically respond to trade barriers through OFDI is critical for sustaining international competitiveness and advancing energy security. This study seeks to address this gap by systematically analyzing firm-level OFDI responses to ADCV shocks in the PV sector, which has been largely overlooked in existing studies. Understanding this linkage has important theoretical implications for the literature on trade barriers and international investment, as well as substantial practical value for guiding policymakers

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Nomenclature abbreviations

ADCV	Anti-dumping and countervailing (double-reverse policy)
OFDI	Outward foreign direct investment
PV	Photovoltaic
GI	Greenfield investment
MA	Mergers and acquisitions

and enterprises in formulating resilient global strategies under external pressure.

OFDI has emerged as a crucial channel for Chinese PV firms to mitigate trade-related risks and sustain global competitiveness. By establishing overseas production facilities or acquiring foreign firms, enterprises can bypass discriminatory tariffs, reduce policy friction, and maintain access to key markets (Amann et al., 2018; Liu et al., 2020; Yuan et al., 2024; Zhao et al., 2024; Liu and Ding, 2023; Lin et al., 2023). Moreover, OFDI enables firms to optimize global resource allocation, acquire advanced technology, and enhance international brand recognition. As a result, OFDI is increasingly viewed not just as a passive consequence of trade friction, but as a proactive strategic response.

Existing studies have extensively examined the impact of trade barriers on export flows, firm performance, and global value chains. Many highlight the damaging effects of ADCV measures on affected firms—through higher tariffs, reduced competitiveness, and disrupted trade relationships (Cheng et al., 2021; Chandra, 2017; Wang et al., 2024; Shen et al., 2021; Lee et al., 2017; Felbermayr and Sandkamp, 2020; Sandkamp, 2020). However, there are still three key limitations in existing research. First, most studies examine trade impacts from a macro or industry-wide perspective, overlooking the heterogeneity of firm-level responses. Second, although some literature on China's PV industry discusses its rise and export competitiveness (Binz and Anadon, 2018; Yao and Cai, 2019; Curley et al., 2023) or the effects of trade restrictions on supply chains and value chains (Chen and Su, 2020, 2022; Teng et al., 2024; Zhang et al., 2025; Yuan et al., 2023; Ding et al., 2019), it rarely investigates how firms strategically adjust their OFDI behavior in response to such policies. Third, the firm-level behavioral mechanisms that link ADCV pressure to international investment are not yet fully understood.

This study seeks to address these gaps by answering three central questions: (1) Do ADCV measures promote OFDI by Chinese PV firms? (2) What are the mechanisms through which these trade barriers influence investment decisions? (3) Does this effect vary by firm characteristics such as size, technological intensity, or export dependence? By exploring these questions, we aim to uncover how trade policy shocks drive cross-border investment behaviors in high-tech manufacturing sectors. Methodologically, we construct a firm-level panel dataset of Chinese listed PV firms spanning 2007 to 2016. Using a multi-period difference-in-differences (DID) strategy, we identify the causal effect of ADCV exposure on OFDI decisions. This empirical strategy enables us to disentangle policy effects from firm-level trends and macroeconomic shocks, ensuring robust inference.

This paper makes three key contributions. First, it introduces a micro-level perspective to the literature on trade policy and firm behavior by leveraging firm-specific data from multiple databases. Second, it identifies multiple mechanisms—such as direct export obstruction, changes in market concentration, and strategic investment shifts—that explain how ADCV pressure affects firm-level OFDI. Third, it explores firm heterogeneity in policy response, offering deeper insight into which types of firms are more likely to adapt via international expansion. Together, these findings not only enrich the literature on trade-investment linkages but also provide actionable insights for enhancing the resilience and global competitiveness of China's PV industry in an increasingly protectionist world.

The remainder of the paper is structured as follows. Section 2 reviews the literature and theoretical framework. Section 3 introduces the empirical model, variable definitions, and data sources. Section 4 presents the estimation results and robustness checks. Section 5 concludes and offers policy implications.

2. Literature review and hypothesis development

2.1. Literature review

Trade barriers, particularly anti-dumping and countervailing (ADCV) measures, are widely used to protect domestic industries from foreign competition. A large body of literature has examined their direct trade effects, showing that such measures suppress export flows by increasing costs and reducing market access for targeted foreign firms (Aggarwal, 2010; Ganguli, 2008; Cheng et al., 2021; Chandra, 2017). For instance, Sandkamp (2020) demonstrated that anti-dumping duties significantly lower import volumes, especially for products facing higher duty rates. Lu et al. (2013) revealed that ADCV measures not only reduce aggregate export volumes but also decrease the number of active exporters. Schiavo et al. (2021) further highlighted the differentiated impacts of anti-dumping actions on export structures, influencing both the extensive and intensive margins. Despite these insights, most studies primarily focus on the macro-level trade-distorting effects, often overlooking firms' adaptive strategies in response to trade barriers.

A second stream of literature has focused on China's photovoltaic (PV) industry, which has emerged as a global leader due to its cost advantages and rapid technological progress (Goodrich et al., 2013; Luo et al., 2021; Shuai et al., 2018). However, external pressures such as ADCV measures imposed by the US and EU have significantly reshaped China's PV export dynamics. Zhu et al. (2021) observed that these trade sanctions altered export market structures and limited firms' expansion capacity. Wang et al. (2020) emphasized the severe disruptions to global PV supply chains caused by trade restrictions. Chen and Su (2022) analyzed supply chain vulnerabilities, while Chen et al. (2017) investigated resource misallocation among Chinese PV exporters following the EU's 2011 ADCV measures. Most of these studies concentrate on export suppression effects, with limited attention to how firms might proactively respond through alternative strategies.

A third group of studies has explored the relationship between trade barriers and outward foreign direct investment (OFDI), highlighting that firms may use OFDI as a strategic tool to bypass protectionist measures and maintain international market presence (Ghodsi, 2020; Cardamone and Scoppola, 2015). However, these studies generally remain at the macro- or industry-level analyses and rarely delve into firm-level behavioral adjustments, particularly in technology-intensive and highly export-dependent sectors like PV. As a result, the dynamics of how ADCV measures specifically motivate Chinese PV firms to undertake OFDI remain underexplored.

Building on these three streams, this paper aims to address these gaps and make distinct contributions. First, unlike most existing research that centers on trade flows or supply chains, this paper investigates how ADCV measures reshape the internationalization strategies of Chinese PV firms at the firm level, focusing on OFDI as a proactive adjustment mechanism. Second, this paper utilizes matched multi-source data to analyze the micro-level strategic responses of PV firms, providing empirical evidence on how trade barriers drive greenfield investments and cross-border mergers and acquisitions. Third, this paper explores the heterogeneity of OFDI responses, examining whether firms' investment choices differ based on their exposure to ADCV measures and firm-specific characteristics. By doing so, this paper not only extends the literature on trade policy and firm behavior but also offers policy-relevant insights for sustaining the global competitiveness of China's PV industry under rising trade protectionism.

2.2. Theoretical analysis and hypotheses

The relationship between trade barriers and outward foreign direct investment (OFDI) has long been debated in the literature. While prior studies primarily focus on macro-level or industry-wide analyses, there is a lack of firm-level evidence, particularly regarding anti-dumping and countervailing (ADCV) measures imposed on Chinese photovoltaic (PV) firms (Tag and Degirmen, 2022; Tekin-Koru and Waldkirch, 2010; Baltagi et al., 2008; Cardamone and Scoppola, 2015). It is generally acknowledged that firms aim to maximize returns and minimize risks when selecting internationalization strategies. Environments with higher economic freedom and institutional efficiency typically offer better expected returns on OFDI, which explains why such countries often become attractive investment destinations (Tag and Degirmen, 2022). In this context, ADCV measures serve as important external shocks that reshape firms' strategic decisions. For example, anti-dumping measures introduced by the EU have been found to stimulate manufacturing OFDI as a means to sustain market access (Cardamone and Scoppola, 2015).

Theoretically, trade barriers can serve as both deterrents and catalysts for OFDI, depending on firm capabilities and market conditions. When firms face high tariffs or non-tariff barriers, they may opt for local production to avoid rising trade costs, thus safeguarding market share and reducing operational uncertainties. This strategic substitution from exports to OFDI is supported by findings that highlight a positive linkage between technical barriers and investment entry modes (Smyth et al., 2017; Ghodsi, 2020). Such barriers transform into incentives for firms to strengthen their global presence rather than relying solely on exports.

Conversely, if trade barriers excessively increase entry costs and operational risks, they may discourage firms from undertaking OFDI altogether. For instance, Duval and Utoktham (2014) emphasize that trade liberalization and improved efficiency significantly boost OFDI outflows, suggesting that harsh protectionist measures can inhibit investment willingness due to high uncertainty and financial burden. In summary, while the impact of trade barriers on OFDI remains controversial in general contexts, Chinese PV firms present a special case. Due to their high export dependence and frequent targeting by ADCV policies, these firms have strong strategic motivations to bypass trade barriers through OFDI. The heightened global protectionism forces them to adopt proactive internationalization strategies to maintain competitiveness and secure market positions (Sun et al., 2020; Ghodsi, 2020).

Moreover, the response to ADCV measures is expected to be heterogeneous across firms. Larger and more productive firms typically possess stronger financial and managerial capabilities, making them more capable of using OFDI as an adaptive strategy, while smaller firms may be forced to reduce operations or exit (Helpman et al., 2004; Antras and Helpman, 2004). Furthermore, ADCV measures often lead to increased market concentration, pushing leading firms to seek overseas expansion as a way to counterbalance shrinking domestic opportunities and to pursue further growth.

In terms of investment modes, firms may select different approaches depending on the severity and urgency of trade barriers. Cross-border mergers and acquisitions (M&A) allow rapid market access and acquisition of strategic assets (Hennart and Park, 1993), which is particularly suitable under strong policy pressures. In contrast, greenfield investment provides long-term operational control and gradual market penetration but requires more time and resources (Slangen and Hennart, 2007).

Based on the above analysis, this study proposes the following hypotheses:

Hypothesis 1. (H1):

Anti-dumping and countervailing (ADCV) measures significantly promote the outward foreign direct investment (OFDI) of Chinese photovoltaic (PV) firms.

Hypothesis 2. (H2):

The positive effect of ADCV measures on OFDI operates through multiple mechanisms:

(a) Trade obstruction pushes firms to switch from export to investment;
(b) ADCV increases market concentration, incentivizing stronger firms to expand overseas to mitigate losses and maintain competitiveness;
(c) Firms respond strategically through greenfield investment or cross-border M&A, with M&A being more efficient under time and cost constraints.

3. Research design

3.1. Sample enterprises and data sources

This study conducts an empirical analysis based on multiple authoritative databases, including the CSMAR Database, the Wind Financial Terminal, the Global Anti-Dumping Database (GAD), the Global Countervailing Database (CVD), the fDi Markets Greenfield Investment Database, and the Zephyr Cross-Border M&A Database. The data on anti-dumping and countervailing (ADCV) measures are collected from the GAD and CVD, which provide detailed records of measures imposed by various countries on Chinese photovoltaic (PV) products. Outward foreign direct investment (OFDI) data are drawn from the fDi Markets and Zephyr databases, covering both greenfield investments and cross-border M&A activities by Chinese firms.

The sample consists of Chinese listed enterprises engaged in the photovoltaic (PV) sector, including manufacturers of solar cells, modules, inverters, and related products, as well as firms with active export activities. These firms are identified through export customs data using HS 6-digit codes such as 854140 (solar cells and modules), 850132 (PV inverters), and 850161 (power conversion devices). This approach ensures that the sample accurately reflects firms operating in the PV industry.

Firms subjected to ADCV measures constitute the treatment group, specifically those producing solar cells and modules that were targeted by ADCV actions during the sample period. In contrast, the control group comprises firms operating in related segments of the PV industry (e.g., inverters and power conversion equipment), whose products were not directly targeted by ADCV measures during the sample period. This design ensures comparability between the two groups while controlling for industry characteristics and mitigating potential selection bias. Firm-level financial and investment data are then matched using unique identifiers from the CSMAR and Wind databases to construct a comprehensive panel dataset.

Representative firms in the treatment group include Trina Solar, Jinko Solar, and JA Solar, whose primary products (solar cells and modules) were subject to ADCV measures in overseas markets such as the EU and the US. In contrast, the control group comprises firms such as Sungrow Power Supply and Ginlong Technologies, which focus primarily on PV inverters and power conversion equipment that were not affected by ADCV measures during the sample period. This categorization illustrates the basis for group assignment and supports the empirical strategy.

The observation period is from 2007 to 2016, selected to capture the most active phase of Chinese PV firms' OFDI expansion and to ensure data completeness and consistency. The final dataset includes 2284 firm-year observations.

3.2. Variable description

3.2.1. Explained variable

This research considers outward foreign direct investment (OFDI) as the key dependent variable. The data processing process is that the time

of greenfield investment is based on the time of the project in the fDi Markets database, and the time of M&A is based on the time of M&A completion in the Zephyr database. Considering that outward investment usually has an extended period, this paper takes whether the enterprise carries out outward direct investment in the next 3 years as an explanatory variable, and the variable is set to 1 if there is an outward direct investment and zero otherwise.

3.2.2. Explanatory variable

The dependent variable is the implementation of anti-dumping and countervailing measures, denoted as "ADCV". The binary dummy variable is an interaction term between time and individuals. It takes 1 to indicate that the firm was affected by PV double-reversal in the year in question and 0 to indicate that it was unaffected.

3.2.3. Control variables

This study also accounts for various factors that could influence firms' OFDI. The following describes the main control variables: (1) Total assets or operating revenues usually measure firm size (Size). Larger firms usually have more resources and substantial market competitiveness and are more likely to invest across borders. (2) Asset turnover (ART): A higher asset turnover ratio usually means that the firm is operating more efficiently and may be more capable and motivated to invest abroad. (3) Age of the firm (lnAge): The firm's age refers to the length of time from its inception to the current time. Mature firms usually have accumulated a wealth of experience and a stable financial position, which makes them more likely to undertake OFDI. (4) Return on Equity (ROE): A higher ROE usually indicates that the firm is more profitable and has more capital to support outward investment. (5) Return on Assets (Earnings): Return on Assets refers to the profitability of a company's assets. A company can realize further growth and expansion through foreign investment when it is more profitable. (6) Nature of Equity (SOE): The nature of equity refers to whether the enterprise is a state-owned enterprise. The government may support SOEs in certain countries or regions and have more tremendous resource advantages, which makes them more likely to make OFDI. (7) Shareholding concentration (Shrcr): Higher equity concentration usually means more efficient decision-making so that firms can make quick decisions when making outward investments. The descriptive statistics of each variable are shown in Table 1.

3.3. Model specification

In this study, to identify the causal effects of ADCV on Chinese PV firms' OFDI, we adopt a multi-period DID approach for empirical analysis. The model is specified as follows.

$$OFDI_{it} = \beta_0 + \beta_1 ADCV_{it} + \sum \beta_i control_{it} + \lambda_i + \delta_t + \mu_{ct} + \pi_{jt} + \epsilon_{icjit} \quad (1)$$

Where i represents firms, t represents year, OFDI denotes outward foreign direct investment by firms, ADCV = treat × post is the interaction term, treat is an individual dummy variable affected by ADCV, and the variable *post* is a time dummy indicating the period following the enforcement of anti-dumping and countervailing measures. HS6 codes are used to identify PV firms subjected to anti-dumping and countervailing investigations. These firms serve as the experimental group and unaffected firms as the control group. Post is 1 if the firm is affected by trade barriers after year t, and 0 otherwise. Control variables include a series of factors that may affect the OFDI of the firms, such as firm size, asset turnover, return on assets, etc., and ϵ_{it} is a random error term.

The DID approach allows us to identify the average treatment effect of ADCV policies by comparing changes in OFDI before and after ADCV enforcement between affected and unaffected firms. Firm, country-year, and product-year fixed effects are included to absorb unobserved heterogeneity, and standard errors are clustered at the firm level to ensure robust inference. A significantly positive coefficient β_1 indicates that ADCV policies promote OFDI by Chinese PV firms, suggesting a proactive outward investment response to trade barriers.

Table 1
Descriptive statistics.

Variable	Obs	Mean	Std. Dev.	Min	Max
OFDI	2284	0.102	0.3027	0	1
ADCV	2284	0.0486	0.2151	0	1
size	2284	7.6273	1.0918	4.4998	12.4525
ART	2284	0.6393	0.3373	0.0895	3.135
lnAge	2284	2.4244	0.5104	0	3.5835
ROE	2284	0.053	0.5011	−20.7367	0.9378
Earnings	2284	0.0606	0.064	−0.5725	0.4172
SOE	2284	0.1532	0.3603	0	1
Shrcr	2284	58.7771	14.3069	15.9534	92.8975
ER	2231	0.1371	0.2163	0	4.13
GI	2284	0.0788	0.2695	0	1
MA	2284	0.035	0.1839	0	1

4. Empirical results and discussion

4.1. Benchmark regression

The regression results from the columns in the table show that ADCV have positively impacted OFDI. Columns (1) and (2) of Table 2 present the regression findings without and with control variables. The estimation results suggest that ADCV coefficients are positive and statistically significant. Columns (3) through (6) of Table 2 further control for industry- and city-year fixed effects and progressively add control variables at each level. The results show that the coefficient of ADCV fluctuates slightly above and below 0.2927, indicating that anti-dumping and anti-subsidy increase the probability of OFDI of Chinese PV firms and that firms make outward investments to circumvent trade restrictions and enhance international competitiveness.

4.2. Robustness tests

4.2.1. Parallel trend test and dynamic test

In analyzing the impact of ADCV on Chinese PV firms' OFDI, this paper refers to Deschenes et al. (2017) to perform a parallel trend analysis. The parallel trend model is an event study design based on the double difference method, which visualizes the dynamic impact of ADCV by estimating the coefficients of time dummy variables in different years before and after the implementation of the policy. The basic form of the model is as follows:

$$OFDI_{it} = \beta_0 + \sum_{s=1}^{4} \beta_{pre_s} D_{pre_s} + \beta_{current} D_{current} + \sum_{s=1}^{5} \beta_{post_s} D_{post_s} + \sum \beta_i control_{it} + \lambda_i + \delta_t + \epsilon_{it} \quad (2)$$

Fig. 1 presents the estimated coefficients of the outward foreign direct investment (OFDI) in different years before and after implementation of ADCV. As can be seen from the figure, before the implementation of the policy (years −4 to −1), the estimated coefficients were close to zero and insignificant, which indicates that before the implementation of the policy, there was no significant difference in the trend of OFDI between the treatment group and the control group. Meanwhile, after the implementation of the policy (years 1–5), the coefficients are positive and gradually increase, reflecting the positive effect of the ADCV on the OFDI of Chinese PV enterprises. In addition, the significance of the estimated coefficients further increases from year 1, indicating that ADCV has a specific cumulative effect.

This result shows that under the pressure of ADCV, Chinese PV enterprises have actively adopted the strategy of outward direct investment to effectively cross the trade barriers by establishing localized production bases or investment and cooperation projects in target

Table 2
Estimation results of the benchmark regression.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	m1	m2	m3	m4	m5	m6
	OFDI	OFDI	OFDI	OFDI	OFDI	OFDI
ADCV	0.1458**	0.1387**	0.2972***	0.2941***	0.2849***	0.2927***
	(0.0668)	(0.0669)	(0.0682)	(0.0646)	(0.0747)	(0.0748)
size		0.0482*		0.0490	0.0463	0.0527
		(0.0274)		(0.0402)	(0.0398)	(0.0389)
ART		−0.0660		0.0224	0.0120	−0.0607
		(0.0478)		(0.0555)	(0.0585)	(0.0683)
lnAge		0.1261**			0.1176**	0.1216***
		(0.0522)			(0.0435)	(0.0437)
ROE		−0.0001			0.0191	−0.0041
		(0.0042)			(0.0163)	(0.0158)
Earnings		0.5472***				0.5645**
		(0.1811)				(0.2397)
SOE		−0.0098				−0.0519
		(0.0244)				(0.0426)
Shrcr		−0.0016				−0.0031***
		(0.0014)				(0.0010)
Constant	0.0949***	−0.4718*	0.0906***	−0.2958	−0.5559*	−0.4133
	(0.0043)	(0.2484)	(0.0037)	(0.2870)	(0.3064)	(0.3397)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	No	No	Yes	Yes	Yes	Yes
City-Year FE	No	No	Yes	Yes	Yes	Yes
Observations	2284	2284	1632	1632	1632	1632
R-squared	0.5477	0.5604	0.7724	0.7740	0.7754	0.7806

Robust standard errors in parentheses.
***p < 0.01, **p < 0.05, *p < 0.1.

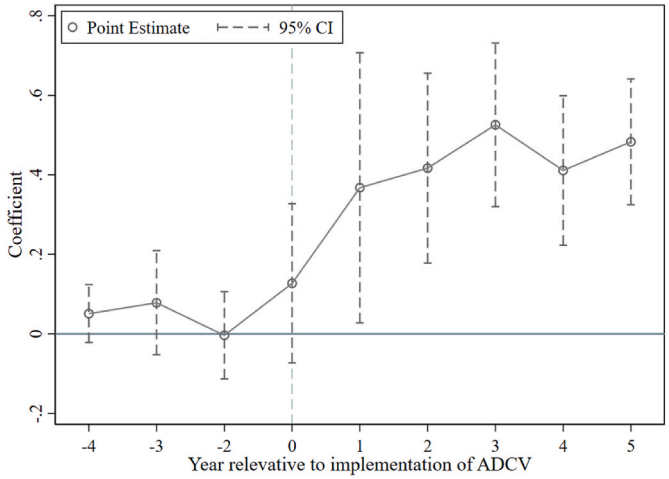


Fig. 1. Parallel trend test and dynamic test.

markets. This way of coping not only avoids the possible adverse effects of the "double reverse" policy but also lays the foundation for enterprises to develop the international market further and enhance the global resource allocation capacity and market competitiveness. At the same time, the gradual strengthening of OFDI also shows that Chinese PV enterprises can flexibly adjust their coping strategies to adapt to the complex international trade environment and realize their sustainable development.

4.2.2. *Placebo test*

To further verify the causal effect of ADCV on Chinese PV firms' OFDI, this paper uses Ferrara et al.'s (2012) placebo test methodology (Ferrara et al., 2012), which tests the causal relationship's robustness by randomizing the policy intervention time. Specifically, the distribution of the coefficients and their significance changes after randomization are observed after 500 times of randomization.

Fig. 2 shows that most of the coefficients after the randomization treatment are concentrated around 0, and most of the estimated coefficients exhibit significant deviations from the actual value of the baseline estimation outcomes, with a predominant leftward distribution. It indicates that both the significance and the intensity of the effect of the policy effect are significantly reduced after the randomization treatment, and the coefficients of the ADCV show outliers in the random sampling pattern, with most of the randomized coefficients below 10 % at the significance level. The above results suggest that random factors do not drive the policy effect's significance but have a strong causal association, thus further validating the robustness of the paper's findings. The placebo test not only excludes the possible chance bias of the model but also strengthens the theoretical and empirical basis for the positive impact of the double anti-dumping policy on the OFDI of Chinese PV enterprises.

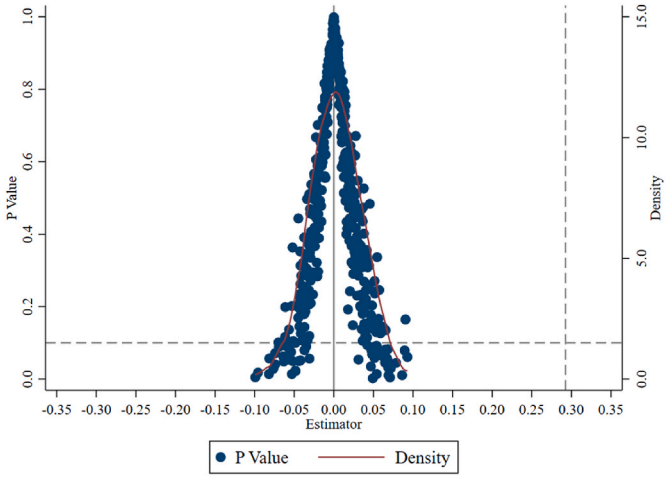


Fig. 2. Placebo test.

4.2.3. Replacement of the measurement model methodology

Given that the explanatory variables in this paper are dummy variables, i.e., binary dependent variables (e.g., whether firms make OFDI), a nonlinear model is used to test the robustness of the regression results further. In this section, we introduce Logit and Probit models, which are suitable for dealing with binary dependent variables and can better capture the nonlinear relationship between explanatory and interpreted variables.

Despite the different estimation methods of Logit and Probit models, the coefficients in columns (1) and (2) of Table 3 remain significantly positive under both models, aligning with the results of the benchmark regression. This result further confirms that the positive impact of anti-dumping and countervailing policies on Chinese PV firms' OFDI is robust and consistent with the findings obtained using linear regression.

4.2.4. Replacement of OFDI measurement

We further conduct a robustness test by substituting alternative indicators for the OFDI variables. Specifically, we use different measures of OFDI to examine the reliability of the results. In addition to using the original OFDI measure, we introduce several other measures, including the number of OFDI (i.e., the number of times a firm has made outward foreign direct investment) and whether a firm has made an outward foreign direct investment within two years and four years. These alternative metrics can measure firms' OFDI behavior from different perspectives and ensure that a single metric does not limit our conclusions. Columns (3), (4), and (5) of Table 3 present the estimation results. Regardless of which new OFDI measure is used, the estimated ADCV coefficients remain significantly positive.

4.2.5. Measurement of ADCV replacement

We adopt an alternative approach to measure the time of implementation of ADCV. In the previous paper, we chose the time of ADCV investigation initiation as the starting point for policy intervention, considering that firms usually take countermeasures before the investigation and that most investigations eventually find the existence of ADCV behavior.

In order to capture the actual impact of anti-dumping countervailing more precisely, this section defines the point in time of the ADCV as the time of the affirmative final determination of the anti-dumping countervailing investigation, i.e., the time of adjudication when a firm is finally found to have engaged in anti-dumping countervailing practices. This point usually marks the formal entry into force of anti-dumping measures, which may have a more immediate and sustained impact. After applying this new criterion to the robustness test, the regression results reveal that the estimated coefficients in column (6) of Table 3 remain significantly positive when using the time of the affirmative final ruling as the policy point in time, consistent with the previous results.

4.2.6. Mitigating sample selection concerns

This study also employs the propensity score-matched double-difference (PSM-DID) method. The core idea of this method is to minimize the effect of selection bias by matching the treatment and comparison groups so that the observable characteristics of the two groups before the policy implementation are as similar as possible. We adopt the cross-sectional and year-by-year PSM-DID methods and use a 1:2 nearest-neighbor matching strategy to ensure that the experimental and control groups have similar characteristics in the pre-policy base period.

The cross-sectional PSM-DID methodology ensures that the experimental and control groups are characterized similarly in the base period by matching the pre-policy cross-section data. We compute the propensity score for each firm and match the appropriate control group sample through a 1:2 nearest-neighbor matching strategy. Table 4 shows that the effect of the ADCV on OFDI is positive, with coefficients of 0.2775 and 0.2767, respectively, suggesting that the policy significantly promotes OFDI by PV firms.

Unlike the cross-sectional PSM-DID, the year-by-year PSM-DID matches each year independently and can better capture changes in policy effects over time. This method can reveal the dynamics of policy effects through year-by-year matching. The regression results of the year-by-year PSM-DID (Column (3) and Column (4) in Table 4) also

Table 3
Other robustness tests.

VAR	(1)	(2)	(3)	(4)	(5)	(6)
	m1	m2	m3	m4	m5	m6
	OFDI	OFDI	OFDI	OFDI2	OFDI4	OFDI
ADCV	1.6818***	3.0871***	0.4624***	0.1453**	0.3575***	
	(0.3150)	(0.6278)	(0.1126)	(0.0696)	(0.0939)	
ADCV1						0.3716***
						(0.0996)
size	0.4750***	0.8562***	0.1063	0.0473	0.0036	0.0545
	(0.1256)	(0.2925)	(0.1217)	(0.0318)	(0.0359)	(0.0391)
ART	−0.4006	−0.8328	−0.1657	−0.0746	−0.0268	−0.0644
	(0.3397)	(0.7258)	(0.1256)	(0.0502)	(0.0487)	(0.0701)
lnAge	−0.1228	−0.1744	0.0953	0.1031**	0.1371**	0.1094**
	(0.2166)	(0.4705)	(0.1434)	(0.0444)	(0.0536)	(0.0435)
ROE	2.1698	5.1707	−0.0485	−0.0697***	0.0004	−0.0039
	(2.1929)	(5.5516)	(0.0324)	(0.0129)	(0.0159)	(0.0154)
Earnings	2.3503	3.2500	1.4017***	0.6438***	0.4230	0.5638**
	(2.8006)	(6.4011)	(0.4110)	(0.1757)	(0.2761)	(0.2360)
SOE	−0.1032	−0.1218	0.0630	−0.0300	−0.0356	−0.0566
	(0.2081)	(0.3959)	(0.1395)	(0.0393)	(0.0374)	(0.0406)
Shrcr	−0.0178**	−0.0328**	−0.0041*	−0.0022*	−0.0033**	−0.0031***
	(0.0073)	(0.0150)	(0.0023)	(0.0012)	(0.0015)	(0.0010)
Constant	2.5981**	11.0585***	−0.6271	−0.3985	−0.0628	−0.3982
	(1.1559)	(1.9552)	(0.8661)	(0.2694)	(0.3691)	(0.3310)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs	1318	1318	1632	1632	1632	1632
R-squared			0.7975	0.7145	0.8135	0.7860

Robust standard errors in parentheses.
***p < 0.01, **p < 0.05, *p < 0.1.

Table 4
PSM-DID.

VARIABLES	(1)	(2)	(3)	(4)
	m1	m2	m3	m4
	OFDI	OFDI	OFDI	OFDI
ADCV	0.2775***	0.2767***	0.2858***	0.2844***
	(0.0784)	(0.0920)	(0.0697)	(0.0738)
size		0.0390		0.0406
		(0.0420)		(0.0400)
ART		−0.0704		−0.0272
		(0.0678)		(0.0682)
lnAge		0.1317**		0.1351***
		(0.0521)		(0.0484)
ROE		0.1221		0.0209
		(0.1956)		(0.0560)
Earnings		0.4381		0.5970*
		(0.4479)		(0.2971)
SOE		−0.0367		−0.0536
		(0.0443)		(0.0458)
Shrcr		−0.0037***		−0.0032***
		(0.0012)		(0.0010)
Constant	0.0915***	−0.2924	0.0896***	−0.3744
	(0.0046)	(0.3994)	(0.0038)	(0.3620)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes
Observations	1318	1318	1586	1586
R-squared	0.7822	0.7888	0.7707	0.7796

Robust standard errors in parentheses.
***p < 0.01, **p < 0.05, *p < 0.1.

show a significant positive effect of ADCV on OFDI, with coefficients of 0.2858 and 0.2844, proving that the conclusions are robust.

4.2.7. Addressing Heterogeneous Treatment Effects

This section aims to uncover potential biases in multi-period DID models, explicitly addressing the effects of heterogeneity in treatment responses. Initially, we apply the diagnostic method De Chaisemartin and d'Haultfoeuille (2020) introduced to evaluate potential biases in the interleaved DID model (De Chaisemartin and d'Haultfoeuille, 2020). The results show that no negative weights are found, suggesting that the multi-temporal DID results are somewhat robust.

To further verify that the heterogeneous treatment effect is insignificant to the results, this paper adopts Sun and Abraham's (2021) method of testing for dynamic effects by using the untreated group as a control group (Sun and Abraham, 2021). This method was chosen because heterogeneous treatment effects may arise from a "bad control group," i.e., a group that started treatment earlier and whose trend has changed as a control group. Fig. 3 shows that the dynamic effects test, considering heterogeneous treatment effects, is consistent with the aforementioned parallel trend test, with no significant differences, indicating that this paper's multi-temporal DID model setup is reasonable.

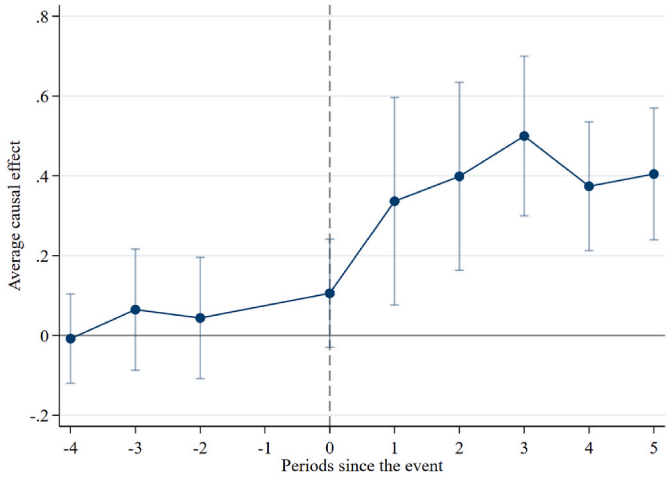


Fig. 3. Addressing heterogeneous treatment effects.

4.2.8. Robustness check using dynamic GMM estimation

To further address potential endogeneity concerns, especially regarding the possible simultaneity between OFDI decisions and ADCV measures, we employ a dynamic system GMM estimator (Arellano and Bover, 1995; Blundell and Bond, 1998). This approach allows us to control for unobserved firm-specific effects and the dynamic nature of OFDI decisions, thereby further strengthening the credibility of our results.

The dynamic GMM results (Table 5) show that the coefficient on ADCV remains positive and statistically significant (0.1242, $p < 0.05$), consistent with our baseline multi-period DID results. The coefficient on the lagged OFDI term is also positive and significant (0.2479, $p < 0.01$), indicating persistence in firms' OFDI decisions. The Hansen test p-value (0.452) indicates that the instruments are valid, with significant AR (1) and non-significant AR (2), thereby supporting the specification validity. These additional results reinforce the robustness of our main conclusion that ADCV measures encourage the OFDI activities of Chinese solar photovoltaic firms.

4.3. Exploring the mechanisms of influence

In this section, we explore the mechanisms by which anti-dumping and countervailing (ADCV) measures affect Chinese PV firms' technological OFDI. Rather than employing the traditional stepwise mediation framework, this study adopts two complementary and theoretically grounded approaches to examine potential mechanisms:

Table 5
Dynamic GMM estimation results.

VARIABLES	(1)
	m1
	OFDI
L.OFDI	0.2479***
	(0.0827)
ADCV	0.1242**
	(0.0552)
size	0.0749***
	(0.0141)
ART	0.0235
	(0.0295)
lnAge	−0.0454**
	(0.0229)
ROE	0.0044
	(0.0073)
Earnings	0.2461
	(0.1566)
SOE	−0.0026
	(0.0236)
Shrcr	−0.0012
	(0.0007)
AR (1)	(0.001)
AR (2)	(0.216)
Hansenp	(0.452)
Constant	−0.3541***
	(0.1028)
Firm FE	Yes
Year FE	Yes
Observations	1982

Standard errors in parentheses.
***p < 0.01, **p < 0.05, *p < 0.1.

(1) We first estimate the effects of ADCV on each mechanism variable (e.g., export share, market concentration, and investment strategy). This verifies whether these channels respond significantly to policy shocks. We then interpret their influence on OFDI primarily through theoretical reasoning and prior empirical literature, rather than introducing them directly as additional mediators in the OFDI regression. This approach corresponds to model (3) in our empirical design and helps avoid potential multicollinearity and over-control issues, while aligning with recent empirical practices (Jiang, 2022).

(2) We further include interaction terms between ADCV and each mechanism variable in the OFDI regression. Significant interaction terms suggest that the marginal effect of ADCV on OFDI depends on these channels, thus providing supporting evidence for the proposed causal pathways. This approach corresponds to model (4) in our empirical design (Acharya et al., 2016).

Specifically, the following three mechanism paths are examined: first, export obstruction prompts PV firms to carry out outward investment to bypass trade barriers; second, firms increase product concentration by focusing on core products to mitigate trade shocks; and third, firms expand into overseas markets to cope with market access barriers through greenfield investment and cross-border M&A strategies.

The empirical strategy relies on two key equations:

$$M_{it} = \beta_0 + \beta_1 ADCV_{it} + \sum \beta_i control_{it} + \lambda_i + \delta_t + \epsilon_{it} \quad (3)$$

$$OFDI_{it} = \beta_0 + \beta_1 ADCV_{it} + \beta_1 ADCV_{it} \times M_{it} + \sum \beta_i control_{it} + \lambda_i + \delta_t + \epsilon_{it} \quad (4)$$

4.3.1. Exports of photovoltaic products hampered by ADCV

From the estimation results in Table 6, ADCV significantly affects both the export share (ER) and outward foreign direct investment of Chinese PV firms. In column (1), the impact of ADCV on export share is negative and significant; the estimated coefficient is −0.0754, indicating that the anti-dumping and countervailing policy will lead to a reduction in the export share of PV products, indicating that the increase in trade barriers hurt exports, ADCV has a "trade destructive effect." In column (2), the ADCV coefficient is 0.4566, indicating that PV firms tend to bypass these trade barriers through OFDI in the face of trade barriers. Further, ADCV × ER coefficient −0.6457 indicates that the reduction of the export share will push enterprises to carry out OFDI and, to a certain extent, verifies the "crossing trade barriers effect" of OFDI.

ADCV limits the ability of Chinese PV firms to export by raising the trade costs of PV products, thus pushing these firms to seek direct investment opportunities in overseas markets. As anti-dumping and countervailing measures expose PV products to higher tariffs or other restrictions in export markets, firms might opt to overcome trade restrictions via OFDI to avoid the adverse effects of such regulations. This strategy helps avoid import tariffs and reduces market entry costs due to trade barriers. Therefore, the empirical results show that the anti-dumping and countervailing policies motivate PV firms to maintain their international competitiveness by strengthening OFDI when they experience export constraints. Further interaction effects show that the reduction in export share exacerbates the increase in OFDI by firms to a certain extent, which also validates the mechanism of the impact of trade barriers on firms' internationalization strategies.

4.3.2. Mitigation mechanisms for market concentration of PV firms

This paper uses the Herfindahl-Hirschman Index (HHI) to measure market concentration (Pavic et al., 2016). In Column (1) of Table 7, the coefficient for ADCV is positive (0.0427), indicating that ADCV has increased market concentration among PV firms. The policy raises barriers for external competitors, enhancing the market share of larger firms. Small and medium-sized enterprises (SMEs) may struggle with higher costs, leading to exits or acquisitions by larger firms, driving industry consolidation. In Column (2), ADCV significantly impacts OFDI (0.3131), but the interaction term (ADCV × HHI) is significantly negative (−1.1690), suggesting that increased market concentration

Table 6
Exports of photovoltaic products hampered by ADCV.

VARIABLES	(1)	(2)
	m1	m2
	ER	OFDI
ADCV	−0.0754**	0.4566***
	(0.0343)	(0.1315)
ADCV × ER		−0.6457***
		(0.2186)
size	−0.0210	0.0465
	(0.0135)	(0.0354)
ART	−0.0094	−0.0678
	(0.0204)	(0.0650)
lnAge	−0.0805	0.1001
	(0.0564)	(0.0769)
ROE	−0.0051	−0.0051
	(0.0035)	(0.0135)
Earnings	−0.0496	0.5212**
	(0.0893)	(0.2205)
SOE	−0.0031	−0.0502
	(0.0105)	(0.0383)
Shrcr	0.0009	−0.0030**
	(0.0006)	(0.0014)
Constant	0.4457***	−0.3126
	(0.1676)	(0.3163)
Firm FE	Yes	Yes
Year FE	Yes	Yes
Industry-Year FE	Yes	Yes
City-Year FE	Yes	Yes
Observations	1580	1580
R-squared	0.9417	0.7812

Robust standard errors in parentheses.
***p < 0.01, **p < 0.05, *p < 0.1.

Table 7
Mitigation Mechanisms for Market Concentration of PV firms.

VARIABLES	(1)	(2)
	m1	m2
	HHI	OFDI
ADCV	0.0427***	0.3131***
	(0.0142)	(0.0884)
ADCV × HHI		−1.1690***
		(0.3310)
size	−0.0262***	0.0455***
	(0.0070)	(0.0170)
ART	0.0048	−0.0657
	(0.0180)	(0.0416)
lnAge	−0.0703***	0.1199***
	(0.0212)	(0.0408)
ROE	−0.0006	0.0006
	(0.0028)	(0.0076)
Earnings	0.0529	0.5501***
	(0.0550)	(0.1311)
SOE	0.0170*	−0.0089
	(0.0087)	(0.0203)
Shrcr	−0.0009***	−0.0015**
	(0.0003)	(0.0008)
Constant	0.6064***	−0.4427***
	(0.0856)	(0.1654)
Firm FE	Yes	Yes
Year FE	Yes	Yes
Industry-Year FE	Yes	Yes
City-Year FE	Yes	Yes
Observations	2260	2260
R-squared	0.7339	0.5635

Robust standard errors in parentheses.
***p < 0.01, **p < 0.05, *p < 0.1.

weakens the effect of ADCV on OFDI. Larger firms, with stronger coping abilities, expand internationally through OFDI, while smaller firms may adopt more conservative strategies and limit OFDI.

4.3.3. The mechanism path of investment strategy selection

According to the empirical results, anti-dumping and countervailing policies promote OFDI by PV firms through GI and MA. In Column (1) and Column (3) of Table 8, the effects of ADCV on greenfield investment and cross-border M&A are both positive and significant, with coefficients of 0.1226 and 0.1969, respectively (both $p < 0.05$). It indicates that ADCV policy significantly promotes firms' expansion into international markets in two ways. Further interaction term analysis shows that the coefficients of the interaction terms of ADCV with greenfield investment and cross-border M&A in columns (2) and (4) are 0.7407 and 0.7587 (both $p < 0.01$), respectively, suggesting that anti-dumping and countervailing policies further strengthen the firms' incentives for OFDI, primarily through greenfield investment and cross-border M&A.

By raising trade barriers, anti-dumping and countervailing policies force firms to resort to greenfield investment and cross-border M&As to circumvent market access barriers and expand into overseas markets. Greenfield investment helps firms to establish new production bases in target markets, which is relatively more of a long-term expansion strategy, while cross-border M&As help to diversify risks and enhance competitiveness by acquiring existing firms to enter the market quickly and gain access to resources. In addition, the push effect of anti-dumping and countervailing policies on cross-border M&As is more significant than that of greenfield investments. In columns (3) and (4), the coefficient of ADCV on cross-border M&A is 0.1969. The interaction term coefficient is 0.7587 ($p < 0.01$), significantly higher than the coefficient of greenfield investment. It suggests that firms are more inclined to enter the market quickly through M&A to cope with the challenge of trade barriers.

Table 8
The mechanism path of investment strategy selection.

VARIABLES	(1)	(2)	(3)	(4)
	m1	m2	m3	m4
	GI	OFDI	MA	OFDI
ADCV	0.1226**	0.1709*	0.1969**	0.1134**
	(0.0455)	(0.1021)	(0.0954)	(0.0466)
ADCV × GI		0.7407***		
		(0.1349)		
ADCV × MA				0.7587***
				(0.0441)
size	0.0476	0.0406	0.0211	0.0490
	(0.0378)	(0.0326)	(0.0241)	(0.0380)
ART	−0.0856*	−0.0725	−0.0092	−0.0514
	(0.0498)	(0.0611)	(0.0826)	(0.0696)
lnAge	0.0794	0.0634	0.0599	0.0980**
	(0.0638)	(0.0715)	(0.0658)	(0.0365)
ROE	−0.0373**	−0.0025	0.0058	−0.0033
	(0.0163)	(0.0133)	(0.0102)	(0.0155)
Earnings	0.6392***	0.4700**	0.1046	0.4972**
	(0.1305)	(0.2194)	(0.2294)	(0.2238)
SOE	−0.0515	−0.0596	0.0144	−0.0536
	(0.0330)	(0.0372)	(0.0245)	(0.0399)
Shrcr	−0.0011	−0.0023*	−0.0026*	−0.0030***
	(0.0013)	(0.0013)	(0.0013)	(0.0009)
Constant	−0.3876	−0.2150	−0.1305	−0.3337
	(0.3283)	(0.2902)	(0.2641)	(0.3024)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes
Observations	1632	1632	1632	1632
R-squared	0.7771	0.7936	0.7031	0.8002

Robust standard errors in parentheses.
***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

4.4. Heterogeneity analyses

4.4.1. Heterogeneity analysis according to the external importance of the Firm's market

This paper measures the importance of external markets based on the ratio of firms' exports to their total operating revenues. When this ratio is high, it indicates that firms are more dependent on external markets. The sample is categorized into firms with high and low external market dependence by applying a median split.

Columns (1) and (2) in Table 9 indicate that the ADCV coefficients for firms with a higher dependence on external markets are more statistically significant. The impact of ADCV is more significant for firms dependent on external markets, and these firms usually prefer to mitigate policy shocks through OFDI when they face policy pressure. Thus, the importance of external markets significantly enhances firms' incentives to respond to trade barriers.

4.4.2. Heterogeneity analysis by firm productivity

Based on the regression results in Columns (3) and (4) of Table 9, we analyze the OFDI responses of high-productivity and low-productivity firms. High-productivity firms (Column 3) exhibit a significant positive coefficient for ADCV (0.4768, $p < 0.01$), suggesting that these firms are more likely to increase OFDI in response to trade policy shocks. In contrast, low-productivity firms (Column 4) display a negative and weakly significant coefficient (−0.0997), indicating a less proactive or even adverse reaction to ADCV measures.

This heterogeneity may be explained by differences in firms' resource allocation efficiency and technological capabilities. High-productivity firms are typically more competitive, with better internal processes and greater capacity to absorb policy shocks. These firms are more agile in reallocating resources and can leverage OFDI to maintain global market presence and competitiveness. In contrast, low-productivity firms often face higher production costs and internal inefficiencies, making them less flexible and more risk-averse under policy pressure. Consequently, they are less inclined—or even unable—to undertake outward investment as a strategic response.

4.4.3. Analysis of regional heterogeneity

Based on the regression results in Column (5) and Column (6) in Table 9, we analyze the OFDI responses of eastern central and western firms. For eastern firms (Column (5)), the coefficient of anti-dumping countervailing policy (ADCV) is 0.3469, indicating that eastern firms actively increase OFDI when facing anti-dumping countervailing policy. Eastern firms have a stronger ability to cope with policy pressure and can circumvent trade barriers through international expansion. As for the central and western enterprises (column (6)), the coefficient of ADCV is −1.3654, which does not show significance, indicating that the central and western enterprises have a weaker response to outward investment when they encounter anti-dumping and countervailing policies, and may even have a negative impact.

This heterogeneity reflects variations in economic development levels., access to resources, and internationalization experience between Eastern central and Western firms. Eastern economies are relatively developed, and firms generally have more substantial market competitiveness, financial liquidity, and internationalization experience. As a result, eastern firms can more effectively utilize policy pressures and adopt OFDI to expand their market share and avoid the risks associated with anti-dumping and countervailing. In addition, Eastern firms are usually located in regions with active foreign trade and have easier access to international markets, thus increasing their global business's stability and growth potential.

In contrast, enterprises in the central and western regions are lagging in accumulating internationalization experience, capital, and resources, facing more tremendous bottlenecks in their development. When confronted with trade barriers, these enterprises often find it difficult to respond quickly through OFDI. Firms in the central and western regions

Table 9
Estimation results of the heterogeneity regression.

VAR	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	m1	m2	m3	m4	m5	m6	m7	m8
	OFDI	OFDI	OFDI	OFDI	OFDI	OFDI	OFDI	OFDI
ADCV	0.1265	0.5075**	0.4768***	−0.0997*	0.3469**	−1.3654	0.3590***	−0.1650
	(0.0768)	(0.2326)	(0.0770)	(0.0536)	(0.1422)	(0.8331)	(0.1080)	(0.2210)
size	−0.0042	0.1098	0.1352**	−0.0159	0.0452	1.6611	0.0226	0.4280*
	(0.0877)	(0.1171)	(0.0590)	(0.0457)	(0.0411)	(1.2114)	(0.0273)	(0.2174)
ART	−0.0056	−0.1536	−0.0787	0.0933	−0.0857	−0.7372	−0.0507	0.0641
	(0.0806)	(0.1343)	(0.0741)	(0.0860)	(0.0615)	(0.7729)	(0.0535)	(0.1989)
lnAge	0.1719	0.1216	0.2169	0.3315	0.1130*	2.0223	0.1497**	0.4781
	(0.1824)	(0.1131)	(0.1328)	(0.2332)	(0.0589)	(1.8312)	(0.0580)	(0.2703)
ROE	0.2241	0.1935	−0.0144	0.0120	0.0019	3.4685	0.0238	0.0029
	(0.4453)	(0.1827)	(0.0130)	(0.0265)	(0.0143)	(2.1918)	(0.0327)	(0.0247)
Earnings	0.0758	1.1069	0.2205	0.0448	0.4699	1.6397	0.4172**	0.2124
	(0.9084)	(1.1046)	(0.4140)	(0.2056)	(0.2958)	(3.6214)	(0.1486)	(0.6236)
SOE	0.0288	0.0273	−0.0297	−0.0671	−0.0111	−1.3480***	0.0126	−0.0339
	(0.0938)	(0.0602)	(0.0726)	(0.0537)	(0.0434)	(0.2530)	(0.0413)	(0.0528)
Shrcr	−0.0050	−0.0040*	−0.0074**	−0.0005	−0.0033**	−0.0514*	−0.0038**	0.0044
	(0.0031)	(0.0023)	(0.0027)	(0.0012)	(0.0013)	(0.0230)	(0.0014)	(0.0037)
Constant	−0.0203	−0.7980	−0.9828**	−0.6730	−0.3136	−15.0949	−0.2508	−4.7368*
	(1.0485)	(0.7553)	(0.4203)	(0.8385)	(0.3445)	(11.1831)	(0.2969)	(2.2012)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	568	577	720	497	1149	65	1257	292
R-squared	0.7993	0.8838	0.8773	0.8140	0.7950	0.9913	0.7700	0.8584

Robust standard errors in parentheses.
***p < 0.01, **p < 0.05, *p < 0.1.

are generally dependent on the domestic market and lack sufficient resource support to deal with the challenges of internationalization, which makes them less capable of responding to anti-dumping and anti-subsidy policies. As a result, the willingness and ability of enterprises in the central and western regions to invest abroad are limited, leading to no significant change in their investment behavior in the face of policy shocks or perhaps even a negative response.

4.4.4. Belt and Road provinces and the OFDI response to trade barriers

To further explore regional policy heterogeneity, we examine whether Chinese solar photovoltaic (PV) firms located in Belt and Road Initiative (BRI) provinces respond differently to anti-dumping and countervailing (ADCV) measures. The BRI is a flagship national strategy aimed at promoting international connectivity and investment by providing selected provinces with preferential policies, infrastructural support, and financial facilitation. We construct a dummy variable to identify firms situated in 18 officially designated BRI provinces (e.g., Guangdong, Zhejiang, Chongqing, Xinjiang) and conduct a subgroup analysis accordingly. The regression results are presented in Table 9 (Columns 7 and 8).

The results indicate that solar PV firms in BRI provinces show a significant positive response to ADCV measures, with a coefficient of 0.3590 (significant at the 1 % level), suggesting a strong tendency to increase outward foreign direct investment (OFDI) as a strategic adaptation to trade barriers. This highlights that the supportive institutional environment and policy incentives in BRI regions strengthen firms' capacity and willingness to expand internationally. In contrast, solar PV firms outside BRI provinces exhibit a negative but statistically insignificant coefficient (−0.1650), implying a limited or even suppressed OFDI response when facing similar trade pressures.

These heterogeneous responses can be attributed to differences in policy support, resource accessibility, and strategic alignment with national priorities. Solar PV firms in BRI provinces benefit from lower transaction costs, stronger external networks, and better financial and infrastructural backing, allowing them to actively pursue overseas opportunities rather than withdrawing from global markets under external shocks. Therefore, our findings demonstrate that the BRI policy environment enables Chinese solar PV firms to actively leverage OFDI as a strategic response to trade restrictions, thereby sustaining their international competitiveness. Moreover, these results underscore the vital influence of regional institutional support and national strategic initiatives in shaping firms' outward investment decisions and enhancing their resilience to external shocks.

5. Conclusion and policy implications

5.1. Conclusion

This research investigates the effect of ADCV on Chinese PV firms' OFDI Using firm-level data from 2007 to 2016. By employing a multi-period difference-in-differences (DID) method, this study provides sufficient empirical evidence to elucidate the relationship. The main conclusions are as follows:

First, anti-dumping and countervailing policies have significantly boosted the OFDI of Chinese PV firms, verifying the existence of the "crossing trade barriers effect." In the face of international trade barriers, enterprises adopt more aggressive investment strategies to avoid tariffs and market access restrictions. The regression analysis results are validated in several robustness tests, indicating that external trade barriers encourage enterprises to enhance their competitiveness in the international market by expanding overseas investment channels.

Second, the mechanism analysis reveals three key ways ADCV promotes OFDI by PV enterprises: First, from the direct mechanism, PV products subject to ADCV are blocked from exporting, prompting PV enterprises to carry out OFDI to cross the trade barriers. Second, from the indirect mitigation mechanism, ADCV investigations drive market concentration in the PV sector, with stronger firms leveraging strategic adjustments to alleviate negative impacts while weaker firms reduce operations or exit. Thirdly, from the perspective of outward investment strategy, anti-dumping and countervailing policies have forced enterprises to adopt both GI and MA by increasing trade barriers to circumvent market access barriers and expand overseas markets. In particular,

photovoltaic enterprises tend to enter overseas markets quickly through shorter cycles and lower-risk M&A to deal with the challenges of trade barriers effectively.

Third, the heterogeneity analysis shows that the facilitating effect of ADCV on OFDI by PV firms is more pronounced among firms with high dependence on external markets, high productivity, located in the eastern region, and situated in Belt and Road Initiative (BRI) provinces. Specifically, these firms typically possess stronger resource integration and market adaptation capabilities, enabling them to react more swiftly to the challenges presented by trade barriers. As a result, they are more inclined to bypass anti-dumping and countervailing measures and expand into overseas markets through OFDI. These findings highlight the differentiated coping strategies and capabilities of different types of firms in the face of external pressures and further reveal the strategic adjustment process of firms under external shocks. The study shows that Chinese PV firms have not been passively retreating due to external trade barriers. Instead, they have enhanced their global competitiveness through active internationalization and overseas expansion, demonstrating their resilience and adaptability in the global trade environment.

5.2. Policy implications

First, strengthen overall policy support frameworks to lay a solid foundation for facilitating and safeguarding Chinese PV firms' internationalization efforts. The government should set up specialized funds to promote cross-border mergers and acquisitions (M&A) and greenfield investments, offer tax incentives, preferential financing, and enhanced political and commercial risk insurance. In addition, it is crucial to strengthen legal and consulting services related to outbound investment, including trade barrier interpretation and international dispute resolution assistance, to lower compliance and operational risks. Meanwhile, developing early warning systems for potential anti-dumping or countervailing investigations can help Chinese PV firms adjust strategies proactively, reduce uncertainty, and mitigate the risk of industrial hollowing-out from large-scale outward relocation.

Second, implement targeted and differentiated policy measures tailored to firm-level and regional heterogeneity to maximize the effectiveness of internationalization strategies among Chinese PV firms. Our findings indicate that PV firms with high external market dependence, high productivity, and those located in eastern and Belt and Road Initiative (BRI) regions are more proactive in using outward foreign direct investment (OFDI) to mitigate trade barriers and strengthen global competitiveness. For these firms, policy priorities should include supporting global supply chain integration, facilitating large-scale overseas capacity deployment, and encouraging cross-border technological cooperation. Specific measures may involve fast-track approval for overseas projects, customized financial instruments, and enhanced export credit insurance. In contrast, PV firms with lower external market dependence, lower productivity, or located in non-BRI and central/western regions face greater challenges; thus, policies should focus on enhancing domestic competitiveness, promoting technological innovation, and improving operational efficiency. Support could include subsidies for technological upgrades, targeted capacity-building programs, and assistance in market diversification.

Third, prioritize enhancing strategic planning and risk management capabilities to ensure the long-term resilience and sustainable international growth of Chinese PV firms. In an increasingly volatile global trade environment, Chinese PV firms need to strengthen their abilities in market access analysis, country-specific risk evaluation, and strategic investment decision-making. Governments can assist by providing standardized risk assessment toolkits, scenario-based training workshops, and comprehensive information platforms to monitor global trade dynamics. Furthermore, actively advancing high-quality development of the Belt and Road Initiative (BRI) and deepening economic and trade cooperation with participating countries can expand international market access and provide broader adjustment spaces for Chinese PV firms, supporting their long-term resilience and sustainable international growth.

5.3. Limitations and future research

While this paper uses data from 2007 to 2016, which may raise concerns about its temporal scope, this period provides substantial analytical value. First, it fully captures the initiation and escalation of anti-dumping and countervailing (ADCV) policies against Chinese PV products, particularly by the U.S. and EU starting in 2011. Second, it coincides with a pivotal phase when Chinese PV firms increasingly turned to outward foreign direct investment (OFDI) to overcome trade barriers and sustain their global market presence. These contextual features make this period particularly suitable for examining the causal impact of trade protection on firm-level internationalization strategies. Nevertheless, it is acknowledged that the global PV industry and international trade dynamics have continued to evolve after 2016, which may affect the generalizability and time sensitivity of the current conclusions. Future research could incorporate extended datasets, when available, to verify whether the "crossing trade barriers" effect persists or transforms over time.

CRediT authorship contribution statement

Boqiang Lin: Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Conceptualization. **Yida Liu:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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